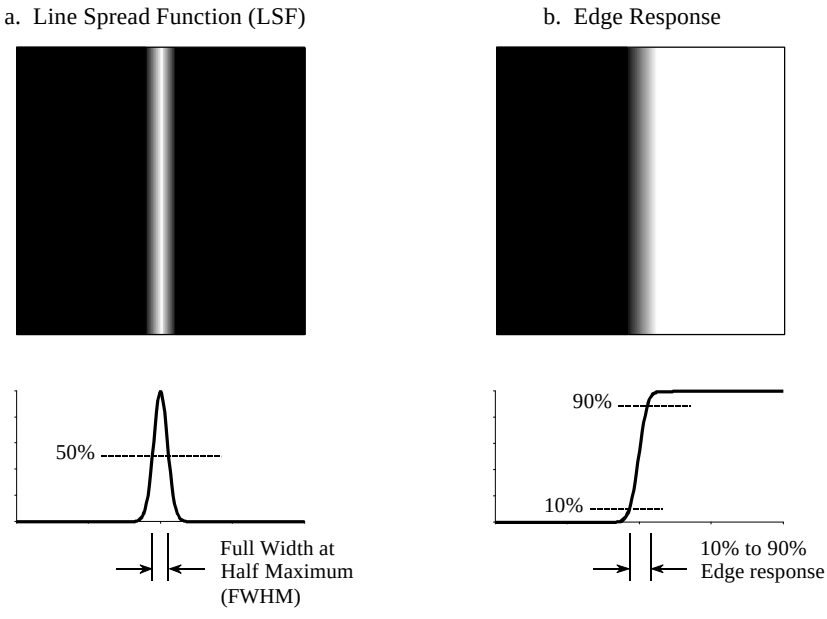

FIGURE 25-3
Line spread function and edge response. The line spread function (LSF) is the derivative of the edge response. The width of the LSF is usually expressed as the Full-Width-at-Half-Maximum (FWHM). The width of the edge response is usually quoted by the 10% to 90% distance.

function is the response of the system to a thin line across the image. Similarly, the edge response is how the system responds to a sharp straight discontinuity (an edge). Since a line is the derivative (or first difference) of an edge, the LSF is the derivative (or first difference) of the edge response. The single parameter measurement used here is the distance required for the edge response to rise from 10% to 90%.

There are many advantages to using the edge response for measuring resolution. First, the measurement is in the same form as the image information is encoded. In fact, the main reason for wanting to know the resolution of a system is to understand how the edges in an image are *blurred*. The second advantage is that the edge response is simple to measure because edges are easy to generate in images. If needed, the LSF can easily be found by taking the first difference of the edge response.

The third advantage is that all common edges responses have a similar shape, even though they may originate from drastically different PSFs. This is shown in Fig. 25-4a, where the edge responses of the pillbox, Gaussian, and exponential PSFs are displayed. Since the shapes are similar, the 10%-90% distance is an excellent single parameter measure of resolution. The fourth advantage is that the MTF can be directly found by taking the one-dimensional FFT of the LSF (unlike the PSF to MTF calculation that must use a two-dimensional Fourier transform). Figure 25-4b shows the MTFs corresponding to the edge responses of (a). In other words, the curves in (a) are converted into the curves in (b) by taking the first difference (to find the LSF), and then taking the FFT.